K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

LINUX BASH SCRIPTING LAB

(ENCS356)

Lab file Submitted to

K.R. MANGALAM UNIVERSITY

For

Bachelor Of Technology

In

Computer Science Engineering Specialization

(Cyber Security)

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Experiment No. 1

Exploring Linux System Commands, Run Levels (Targets), and Help Utilities.

Problem Statement

Efficient use of the Linux operating system requires a strong understanding of basic system commands, system states, and built-in help mechanisms. Unlike graphical operating systems, Linux relies heavily on command-line utilities for system interaction, administration, and automation.

The problem is to explore and practice essential Linux system commands, understand system run levels / targets, and effectively use Linux help utilities such as man, *--help*, and *auto*- completion to access command documentation and usage information.

Aim

To study and execute basic Linux system commands, understand Linux run levels (systemd targets), and use Linux help utilities such as `man`, `--help`, and tab auto-completion.

Theory

Linux Command Line Interface (CLI)

The Linux Command Line Interface allows users to interact with the operating system by typing commands. It provides better control, flexibility, and efficiency compared to graphical interfaces, especially for system administration and automation.

System Run Levels / Targets

- Modern Linux systems use **systemd**, where traditional run levels are replaced by **targets**.

Common systemd targets:

- `graphical.target` – Multi-user mode with GUI
- `multi-user.target` – Multi-user mode without GUI
- `rescue.target` – Single-user rescue mode
- `poweroff.target` – Shutdown system
- `reboot.target` – Reboot system

Linux Help Utilities

Linux provides built-in help tools for learning commands:

- `man` – Displays complete manual pages
- `--help` – Shows brief command usage
- **Tab auto-completion** – Completes commands and file names automatically

Commands Used

Commands	Description
`pwd`	Display present working directory
`ls`	List files and directories
`cd`	Change directory
`whoami`	Display current user
`date`	Show system date and time
`uptime`	Show system running time
`man`	Display manual pages
`--help`	Display help for a command
`systemctl`	Manage system services and targets
`runlevel`	Display previous and current run level
`clear`	Clear terminal screen

Procedure / Steps

Step 1: Login:

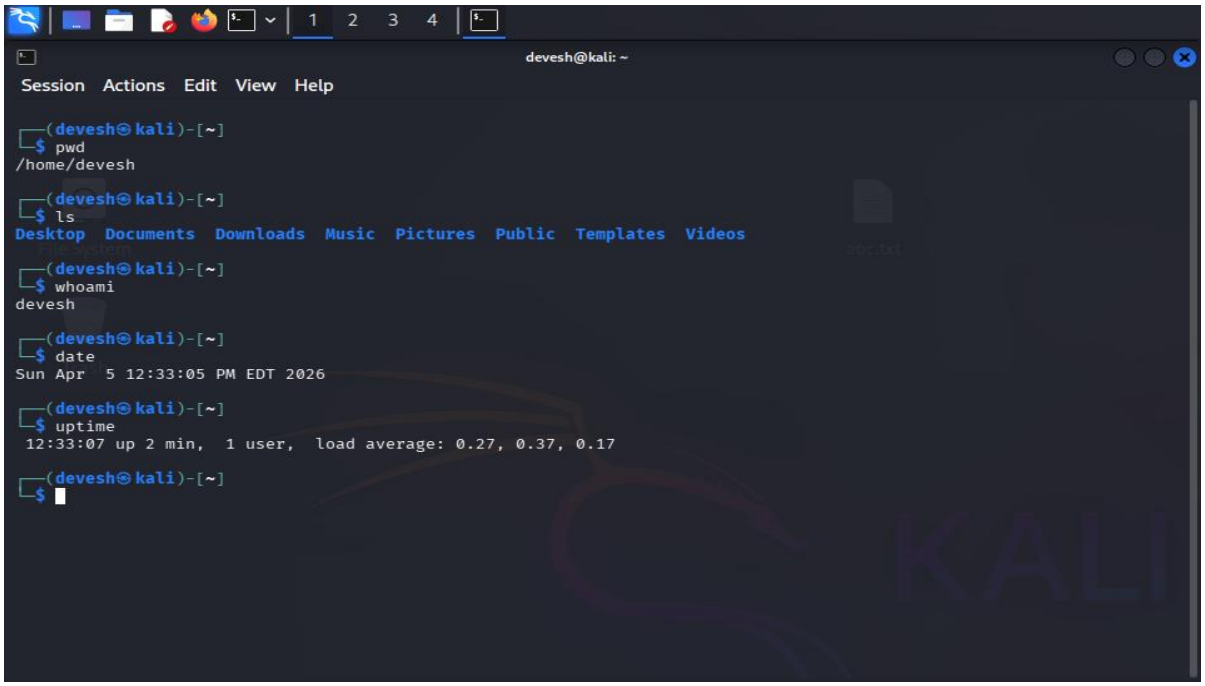
Log in to the Linux virtual machine and open the Terminal.

Step 2: Execute Basic Linux Commands in the Terminal.

Commands:

- `pwd` shows current directory
- `ls` lists files
- `whoami` displays logged-in user
- `date` shows system date/time

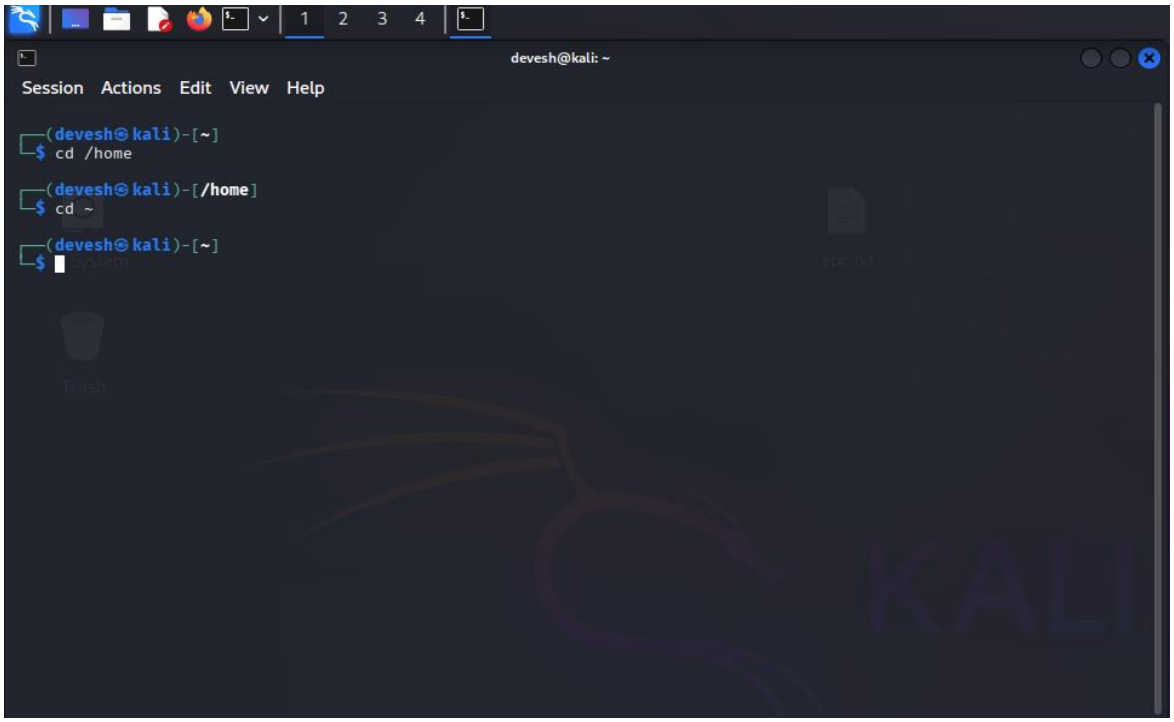
- `uptime` shows how long system is running



Step 3: Directory Navigation using:

- `cd /home` moves to home directory.

- `cd ~` returns to user's home directory.

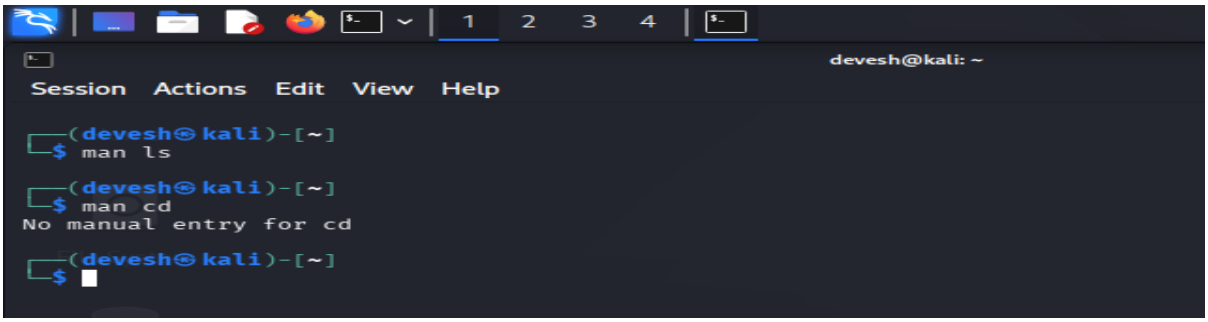


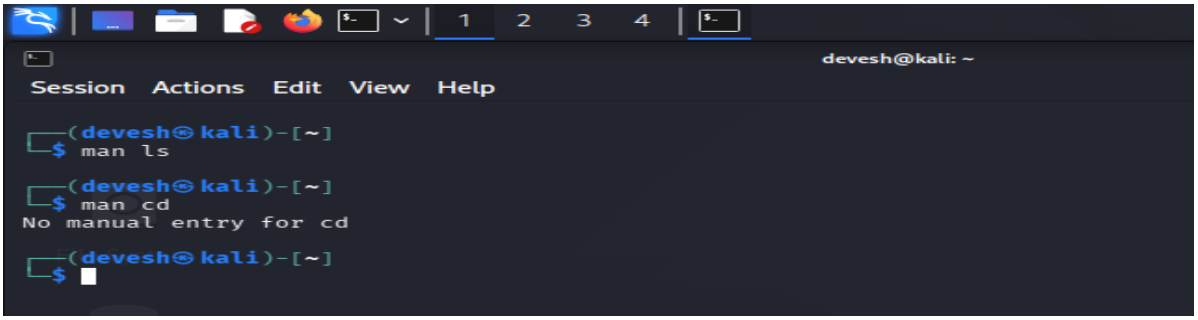
Step 4: Using Manual Pages

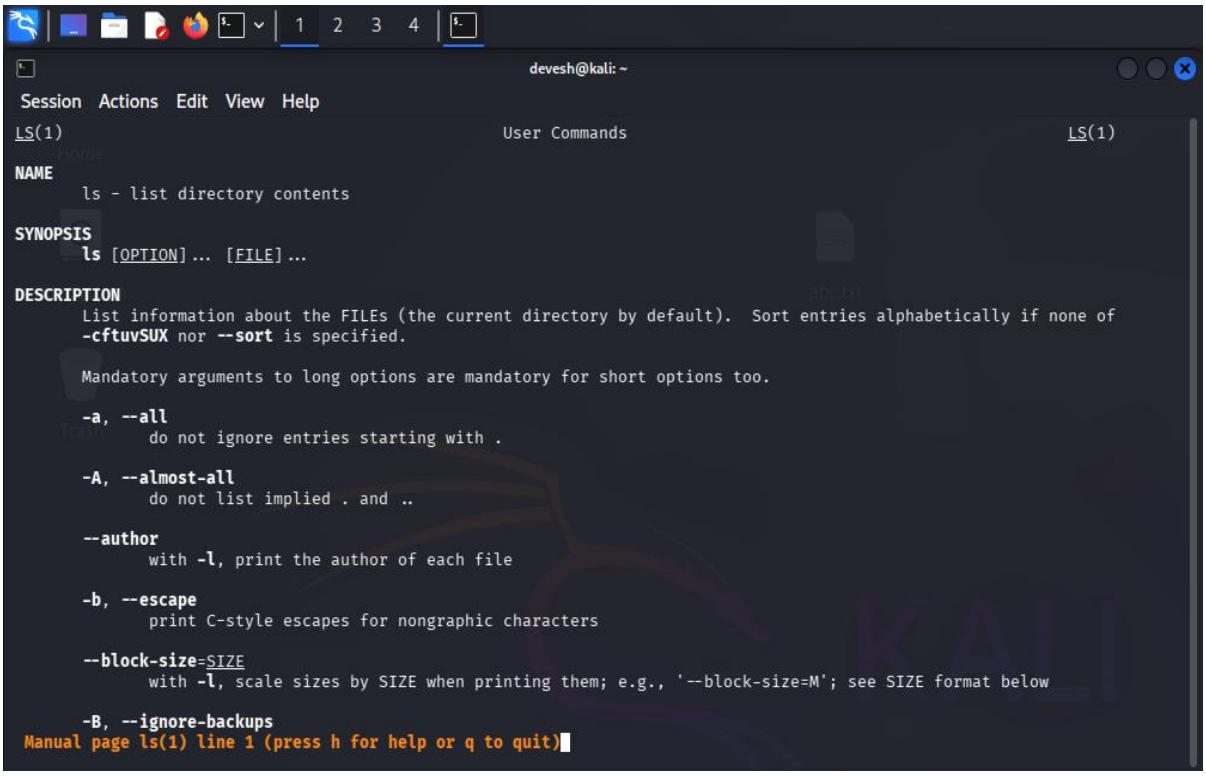
- `man ls`
- `man cd`

It Read command details.

➢ Press `q` to exit the manual.







Step 5: Using Help Option:

- `ls -help`
- `systemctl -help`

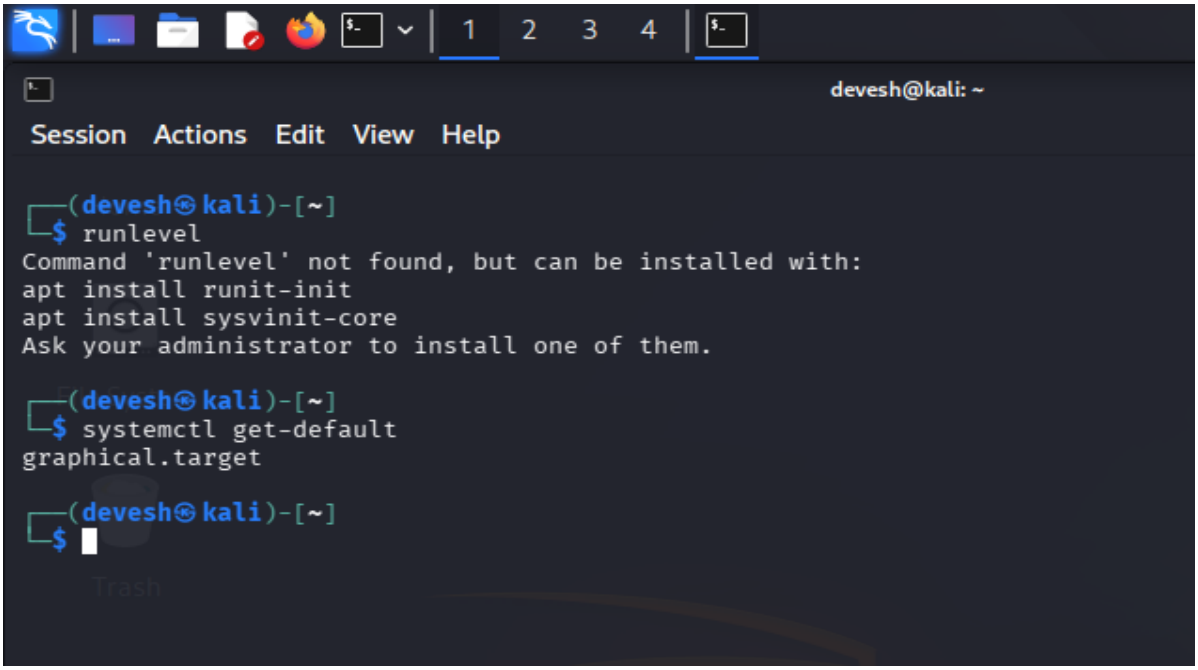
This displays short usage instructions and option.

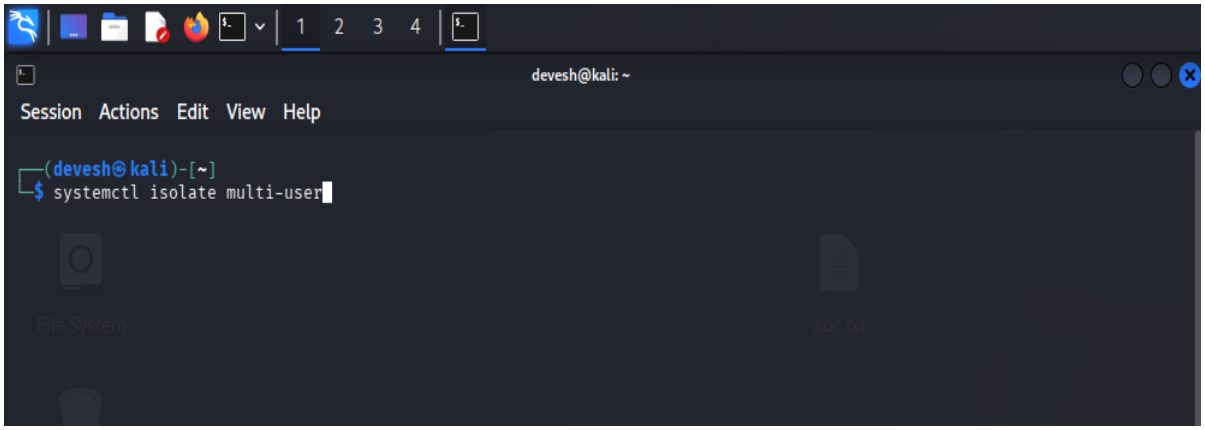
```
devesh@kali: ~
Session  Actions  Edit  View  Help

┌──(devesh㉿kali)-[~]
└─$ man cd
No manual entry for cd

┌──(devesh㉿kali)-[~]
└─$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                  do not ignore entries starting with .
  -A, --almost-all           do not list implied . and ..
      --author               with -l, print the author of each file
  -b, --escape               print C-style escapes for nongraphic characters
      --block-size=SIZE      with -l, scale sizes by SIZE when printing them;
                             e.g., '--block-size=M'; see SIZE format below

  -B, --ignore-backups       do not list implied entries ending with ~
  -c                         with -lt: sort by, and show, ctime (time of last
                             change of file status information);
                             with -l: show ctime and sort by name;
                             otherwise: sort by ctime, newest first

  -C                         list entries by columns
      --color[=WHEN]         color the output WHEN; more info below
  -d, --directory            list directories themselves, not their contents
  -D, --dired                generate output designed for Emacs' dired mode
  -f                         same as -a -U
  -F, --classify[=WHEN]      append indicator (one of */=>@|) to entries WHEN
```

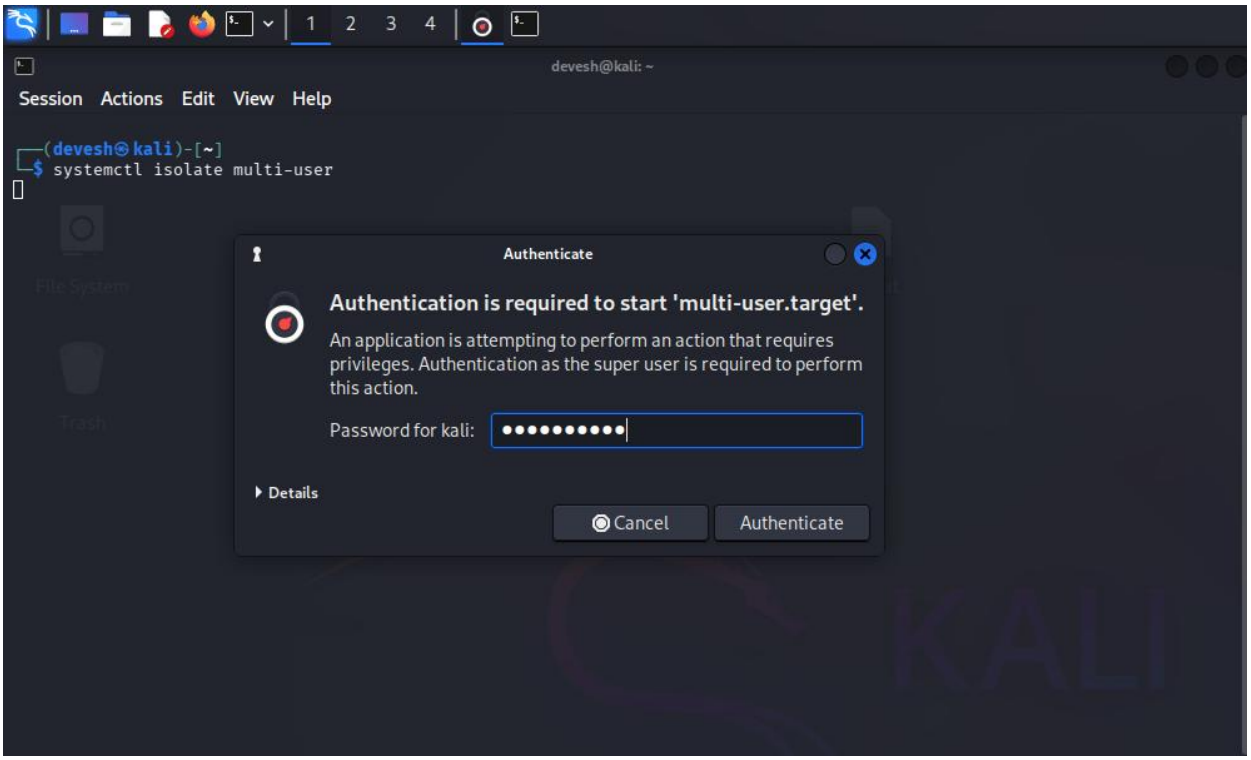
```
devesh@kali: ~
Session  Actions  Edit  View  Help

┌──(devesh㉿kali)-[~]
└─$ systemctl --help
systemctl [OPTIONS...] COMMAND ...

Query or send control commands to the system manager.

Unit Commands:
  list-units [PATTERN...]             List units currently in memory
  list-automounts [PATTERN...]        List automount units currently in memory,
                                      ordered by path
  list-paths [PATTERN...]             List path units currently in memory,
                                      ordered by path
  list-sockets [PATTERN...]           List socket units currently in memory,
                                      ordered by address
  list-timers [PATTERN...]            List timer units currently in memory,
                                      ordered by next elapse
  is-active PATTERN...                Check whether units are active
  is-failed [PATTERN...]              Check whether units are failed or
                                      system is in degraded state
  status [PATTERN...|PID...]          Show runtime status of one or more units
  show [PATTERN...|JOB...]            Show properties of one or more
                                      units/jobs or the manager
  cat PATTERN...                      Show files and drop-ins of specified units
  help PATTERN...|PID...              Show manual for one or more units
  list-dependencies [UNIT...]         Recursively show units which are required
                                      or wanted by the units or by which those
                                      units are required or wanted
  start UNIT...                       Start (activate) one or more units
  stop UNIT...                        Stop (deactivate) one or more units
  reload UNIT...                      Reload one or more units
```

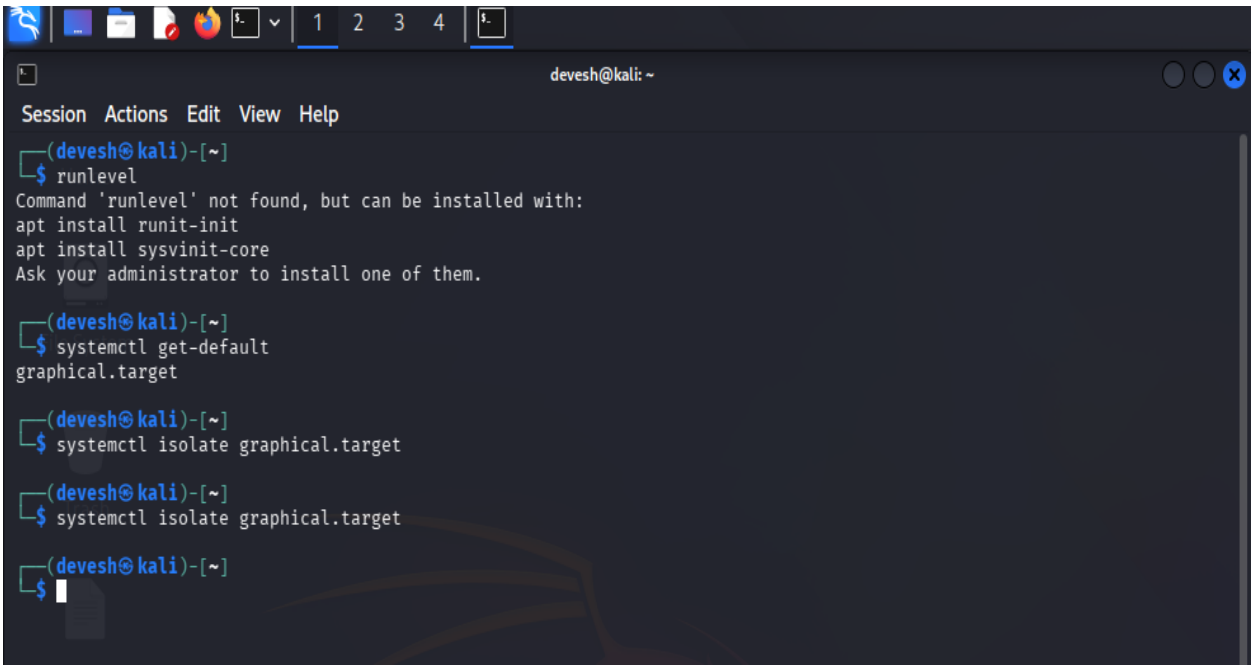
Step 6: Check System Run Level / Target

- `runlevel`- "Kali Linux uses systemd, so traditional runlevel commands are deprecated and `systemctl` targets are used instead."
- `systemctl get-default` shows default system target.






1 2 3 4
devesh@kali: ~
Session Actions Edit View Help
┌──(devesh㉿kali)-[~]
└─$ systemctl isolate multi-user
Authenticate
Authentication is required to start 'multi-user.target'.
An application is attempting to perform an action that requires privileges. Authentication as the super user is required to perform this action.
Password for kali:
Details
Cancel
Authenticate


1 2 3 4
devesh@kali: ~
Session Actions Edit View Help
┌──(devesh㉿kali)-[~]
└─$ runlevel
Command 'runlevel' not found, but can be installed with:
apt install runit-init
apt install sysvinit-core
Ask your administrator to install one of them.
┌──(devesh㉿kali)-[~]
└─$ systemctl get-default
graphical.target
┌──(devesh㉿kali)-[~]
└─$ systemctl isolate graphical.target
┌──(devesh㉿kali)-[~]
└─$ systemctl isolate graphical.target
┌──(devesh㉿kali)-[~]
└─$

Step 7:- Practice auto-completion by typing partial command names and pressing Tab.

Step 8 :- Clear the terminal screen: Clear

Outcomes

- Successful execution of basic Linux commands
- Display of manual pages and help options
- Identification of current run level/target
- Improved command-line efficiency

Result

Thus, basic Linux system commands, system run levels, and help utilities were successfully explored. The student gained practical understanding of Linux command-line interaction and Linux documentation tools.